

Protocol for leaf-level gas exchange measurement for photosynthesis model calibration

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1 Background

Reliable leaf-level measurements of photosynthesis are critical for calibrating biochemical and empirical models of carbon assimilation, which support crop simulation, land surface modeling, phenomics, and ecophysiology. In turn, these models help to inform predictions across scales including seasonal crop yields and decadal atmospheric CO₂ trends. However, available data and published model parameters for analysis and forward prediction remain sparse and inconsistent (Rogers, 2014), and a detailed, standardized protocol focused on efficient gas exchange data collection for photosynthetic model calibration is absent. Establishing such a protocol could accelerate progress in both climate modeling (Rogers et al., 2017) and crop improvement (Bailey-Serres et al., 2019) by providing a consistent foundation for model calibration and comparison.

The most widely used model of C₃ photosynthesis is the Farquhar, von Caemmerer, Berry (FvCB) biochemical model (Farquhar et al., 1980), often extended with a triose-phosphate utilization (TPU) rate limiting state (Harley and Sharkey, 1991), peaked temperature responses (Bernacchi et al., 2003; Medlyn et al., 2002), and a more flexible light response function (Farquhar and Wong, 1984). The model predicts the net carbon assimilation rate of photosynthesis (A) as a function of internal CO₂ concentration (C_i), absorbed photosynthetically active radiation (Q), and leaf temperature (T_l), parameterized with dozens of genotype- and environment-dependent parameters. This model and its numerous variants have been widely applied across the plant sciences, from molecular biology to agricultural decision support (e.g., Feyissa et al., 2019; Muller and Martre, 2019), and integrated into large Earth systems models (Rogers et al., 2017) used to investigate future climate scenarios. A major bottleneck in model utilization is obtaining data of sufficient quality and quantity for robust model calibration for a given set of plants in question.

For decades, portable gas exchange systems equipped with infrared gas analyzers (IRGAs) have enabled precise quantification of CO₂ and H₂O fluxes at the leaf level (Busch et al., 2024). These gas exchange measurements paired with those of environmental conditions provide the means for deriving key photosynthetic parameters. However, many experimental designs used throughout the literature to obtain environmental response curve data vary in approach and often fall short of supporting full photosynthetic model parameterization (Rogers et al., 2017). Furthermore, calibrated parameters are highly sensitive to fitting techniques (Long and Bernacchi, 2003; Gu et al., 2010) and model formulations (Lochocki and McGrath, 2025). These challenges result in published parameter sets that can be incomplete, inconsistent, and incomparable across studies.

Traditional approaches for estimating photosynthetic parameters often perform steady-state responses of A to C_i (known colloquially as “ A - C_i ” curves), as well as to light (A - Q) and temperature (A - T). Measurements are time consuming, and each environmental response curve is typically generated independently and fit independently instead of together, as one unified set of measurements for model parameter extraction. Recent advances in the speed of A - C_i curve measurement (Saathoff and Welles, 2021; Tejera-Nieves et al., 2024) and in robust FvCB model fitting (Lei et al., 2025), have increased the throughput of reliable model calibration.

Yet another challenge in obtaining data for calibrating photosynthetic models is the trade-off between measurement resolution and canopy representation. Photosynthetic parameters can vary substantially from

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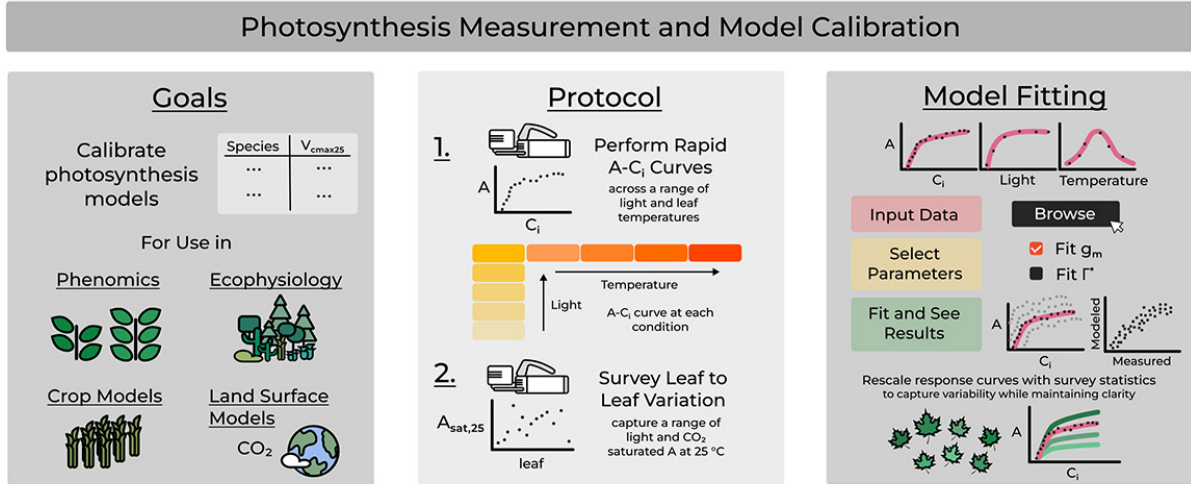


Figure 1: Schematic depiction of a protocol for calibrating leaf-level models of photosynthetic assimilation (A) with respect to internal CO₂ concentration (C_i), absorbed light flux, and leaf temperature. Rapid A - C_i curves are performed on one or two leaves across different temperatures and light levels. Survey measurements at saturating light and CO₂ and at a common reference temperature are then taken to capture the natural leaf-to-leaf variability. The survey measurements are used to rescale the A - C_i curves performed on a single consistent leaf to better reflect the average of the canopy. Model fitting is performed using PhoTorch Studio, a graphical user interface for the PhoTorch fitting package (Lei et al., 2025).

leaf to leaf, often driven by differences in age and nutrient content – particularly nitrogen and phosphorous – linked to position in the canopy (Field, 1983; Wilson et al., 2000; Schultz, 2003; Walker et al., 2014; Bloomfield et al., 2018). However, conducting the full suite of photosynthetic response curves across CO₂, light, and temperature, at a sufficient resolution to extract key traits, such as the temperature optimum of photosynthesis, is time intensive, rendering comprehensive canopy coverage impractical. An alternative approach is to conduct point-wise survey measurements across many leaves, leveraging ambient environmental variation to generate canopy-scale “response curves”. While more sample efficient, this approach introduces substantial noise that obscures the nonlinear shape of physiological responses, complicating parameter estimation and interpretation within models based on mechanistic, leaf-level processes.

To address these challenges, we present a semi-automated protocol for measuring leaf-level gas exchange for photosynthetic model calibration that attempts to balance quality and efficiency. The protocol utilizes rapid A - C_i response curves based on the Dynamic Assimilation™ Technique of the LI-6800 Portable Photosynthesis System (LI-COR Biosciences, Lincoln, NE, USA) across a range of light and temperature values, as well as survey measurements to capture leaf-to-leaf variation in the magnitude of A under saturating conditions. FvCB model calibration is demonstrated using a drag-and-drop user interface that provides simple parameter extraction and quick result visualization. Approximately 60% of the protocol can be automated with scripts that set instrument cuvette conditions, run the response curves, and log the data. The end-to-end solution extracts 24 possible FvCB parameters with a single portable gas exchange system in 2.5 hours. By standardizing and automating response curve measurements and streamlining model calibration for users of all technical ability, this protocol provides a means for reliable, reproducible, and efficient extraction of photosynthetic parameter values to support the growing demand in the photosynthesis modeling community.

2 Materials and conditions

2.1 Ambient conditions checklist

This protocol is intended to be applied on plants in the field, and certain ambient weather conditions will limit its feasibility. Model calibration will be most effective with ambient light reaching 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ and leaf temperatures able to range from 25 °C to 43 °C. This can most consistently be achieved on a day with:

1. Full sun and no or minimal cloud coverage
2. Minimal wind
3. A daily high air temperature of at least 18°C (65°F)

A rule of thumb to keep in mind is the LI-6800 chamber conditions can achieve roughly $\pm 10^{\circ}\text{C}$ from the console temperature, and console temperature can reach 5-15°C above ambient air temperature when heated in full sun.

Preferred conditions for performing this protocol typically occur between mid-morning and solar noon, when gas exchange rates are naturally near a peak, water potentials are not too low, and a range of cuvette temperatures can be reached.

This protocol could be adapted to perform measurements in the laboratory, but this will likely limit the range of possible environmental conditions. In this case, it would be desirable to measure plants growing in the brightest light and warmest air temperature possible.

2.2 Plant materials

This protocol was developed to be used on intact, fully mature, healthy leaves of broadleaf plant species. The underlying photosynthetic parameters extracted from the data are expected to exhibit seasonal, phenological, and interannual change, so a modeler should be aware of when calibration data was collected and for what period in time model parameters will be used in simulation or for comparison.

3 Equipment

Specific instrument settings described in this protocol are for the LI-COR instrument listed below. Adaptations may be needed in order to use instruments from other manufacturers.

1. LI-6800 Portable Photosynthesis System (LI-COR Biosciences, Lincoln, NE, USA)

4 Software

The protocol can be semi-automated with Background Programs written for the LI-6800. Three programs are provided, one that carries out the light sweep, one that carries out the temperature sweep, and one that carries out light sweep followed by the temperature sweep. The protocol can also be applied without the automation script by manually varying each environmental setpoint and running the CO₂ response program already included in the instrument firmware.

In addition to the data collection, we also discuss parameter calibration based on PhoTorch (Lei et al., 2025).

List of optional software:

1. LI-6800 Background Programs for automated measurements of light and temperature sweeps (<https://github.com/PlantSimulationLab/LI6800-automated-scripts>)
2. PhoTorch (<https://github.com/GEMINI-Breeding/photorch>), for advanced users looking to automate workflows of fitting large datasets with Python
3. PhoTorch Studio (<https://github.com/ktrizzo/photorch-studio>), for simple drag-and-drop fitting with a graphical user interface

To use the automated Background Program scripts, first transfer them into the LI-6800's BP Directory.

1. Download the .py script files (e.g., `ACi_Light_Sweep.py`) from the `automated-scripts` repository above, and transfer to a USB thumb-drive.

Table 1: $A-C_i$ response curve light and temperature combinations. Each numbered entry represents an $A-C_i$ curve measurement, with its position on the grid denoting the light–temperature setting. Measurements 1-6 (colored in red) are performed on the same leaf, and 7-12 (colored in blue) performed on a different second leaf. Maximum achievable temperatures will depend on ambient conditions. ‘S’ denotes the survey measurements, which should each be performed on a different leaf.

Light ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	Air temperature ($^{\circ}\text{C}$)					
	25	30	35	38	40	43
2000	1/7/S	8	9	10	11	12
1600	2					
1200	3					
600	4					
200	5					
0	6					

Table 2: Approximate time requirements for protocol tasks. Setup and instrument calibration are needed once daily. The light and temperature component time estimates are based on completion with the automated scripts – some additional time would be needed for manual setting of controls. Completing the protocol yields a complete set of average FvCB parameters for the species, including CO_2 , light, and temperature response.

Protocol Tasks	Approximate Time (min)
Setup and Instrument Calibration	30
Light Sweep	40
Temperature Sweep	50
Survey	30
Model Fitting	1
Total	151 (~ 2.5 hrs)

2. Connect the LI-6800 head to the console (file access will not be available otherwise) and turn the machine on. Plug the USB thumb-drive into the LI-6800.
3. Navigate to the LI-6800 file browser under ‘Tools’ $\rightarrow$ ‘Manage Files’. Select ‘Copy files to LI-6800’, and under the ‘Copy to’ drop-down, select ‘BP Directory’. Find the protocol .py files, select them and press ‘Copy’ to transfer the files to the device. Safely eject the USB with ‘Eject’.
4. (**Important**) Add user variables under the ‘Constants’ $\rightarrow$ ‘User’ tab. Press ‘Add’ followed by ‘Edit’ and rename the variable **CurveID**. Make another variable and rename it **Response**. This step is necessary to log curve ID’s throughout the protocol.

5 Procedure

5.1 Scenarios

Depending on the number of LI-6800 instruments available, two sampling scenarios are proposed. The environmental conditions for the measurement procedure are shown graphically in Table 1.

1. **Scenario 1: One LI-6800.**

- Perform the light-sweep measurements 1–6 of Table 1 on a single healthy, sunlit leaf (~ 40 min).
- Perform the temperature-sweep measurements 7–12 of Table 1 (or as hot as achievable) on the same leaf (~ 50 min).
- Then perform survey measurements on 15–25 diverse leaves at $T_{\text{air}} = 25^\circ\text{C}$, $Q = 2000 \mu\text{mol m}^{-2} \text{s}^{-1}$, $C_a = 2000 \mu\text{mol mol}^{-1}$ (~ 30 min).

2. Scenario 2: Two LI-6800 instruments.

- Instrument A collects measurements 1–6 of Table 1 on a single healthy, sunlit leaf while instrument B simultaneously collects 7–12 of Table 1 on a separate neighboring leaf.
- When instrument A finishes first, perform the survey measurements on 15–25 leaves at the survey set-points above.

5.2 LI-6800 initial setup

Prior to use in respective protocols, the chosen instruments will need to be set up, typically once per day or per measurement session. The set-up procedure described is not meant to be exhaustive and represents only typical set-up tasks performed before each measurement session. Users should consult manufacturer documentation for a complete guide on instrument set-up.

1. Instrument set-up and power on

- Connect head (hand-held unit) via cable and tube.
- Check the head has correct aperture size installed. The largest aperture should be used that can be consistently covered *entirely* by sampled leaves.
- Connect H_2O add column, and ensure there is plenty of water in the column.
- Install new CO_2 cartridge.
- Check the desiccant level. If there is less than 1/2 orange Sorbead[®] (or blue Drierite[®]) left, replace it. Note that it may be possible to replace only the half of the column that has lost its color.
- Remove red cover from light sensor.
- Power on instrument.

2. Aperture size: On the ‘Start up’ $\rightarrow$ ‘Chamber Setup’ tab, click the button for the appropriate aperture size that was installed in the previous step above.
3. Dynamic Equations: On the ‘Start up’ $\rightarrow$ ‘Chamber Setup’ tab, ensure ‘Add Dynamic Equations’ is checked.
4. Clock Synchronization: On the ‘Start up’ $\rightarrow$ ‘System Setting’ $\rightarrow$ ‘Date Time & Clocks’, press ‘Sync Head Clock’.
5. Then, on the ‘Constants’ $\rightarrow$ ‘Gas Exchange’ tab, check ‘UseDynamic’. (If the $A-C_i$ curves look very off, not having this box checked could be the issue; this can be fixed post-facto by changing ‘UseDynamic’ from FALSE to TRUE in the exported .xlsx file.)

5.3 LI-6800 calibration

5.3.1 Set Environment

Close the chamber and leave it empty. Go to the ‘Environment’ tab and set the following environmental parameters.

5.3.2 Matching and Tuning

1. Perform CO_2 and H_2O point matching: on the ‘Measurements’ tab, click ‘Match IRGAs’. In the upper right, click ‘Auto’ under ‘Matching at a single reference point’. When both the CO_2 and H_2O lights turn green simultaneously, matching is complete. Double-check the ‘Time Since Last Match’ field to verify success.

Table 3: Instrument settings for LI-6800 (LI-COR Biosciences, Lincoln, NE, USA) to be used for initial setup and daily calibrations.

Group	Parameter	Value
Flow	Flow	Auto – 500 $\mu\text{mol s}^{-1}$
	ΔP	Auto – 0.1 kPa
	Pump (speed)	Auto
H ₂ O	RH _{air}	50%
CO ₂	CO ₂ _r	400 $\mu\text{mol mol}^{-1}$
	Soda Lime – Scrub	Auto
Fan	Fan Speed	10 000 rpm
Temperature	Temperature	Off
Light	Head Light Source Setpoint	2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$

2. Perform CO₂ and H₂O range matching¹: ‘Constants’ → ‘Range Match’. Click ‘CO₂ Range Match’ → ‘Start’. Select the Normal (5 min) match and set Flow_s/Flow_r = 1.13. Repeat for H₂O, selecting the Quick (3 min) match.
3. Perform CO₂ and H₂O dynamic tuning: ‘Constants’ → ‘Dynamic’. Ensure CO₂ and H₂O control are on. ‘Utilities/Tests’ tab → choose ‘CO₂ Test Current’ → ‘Start’. After 2 min, select ‘Yes’ to store. Repeat for ‘H₂O Test Current’.

5.4 A-Ci Curves

For each sample temperature and light combination in Table 1, collect an $A-C_i$ curve measurement. Instructions are first given for manual setting of chamber conditions for temperature and light sweeps, followed by the procedure when using automated scripts to sweep through environmental conditions.

(Highly recommended) In the ‘Measurements’ tab, graph A versus C_i to observe each $A-C_i$ curve in real time. Curves should appear as increasing hyperbolas that saturate at high CO₂ (e.g., Fig. 2); deviations may indicate instrument or leaf issues.

1. Choose a fully-expanded, outer-canopy sunlit leaf that is large and appears healthy. Clamp the chamber onto the leaf, ensuring it covers the entire aperture.
2. Create a folder with the crop name and any desired identifiers.
3. (Manual Option) **Open a log file**: ‘Log Setup’ → ‘Open a Log File’. Use the naming convention: {YYYY-MM-DD}_{time}_Q{light}_T{temperature}.
4. (Manual Option) Set the appropriate air temperature: ‘Environment’ → ‘Temperature’. Use $T_{\text{air}} = 25, 30, 35$ °C, etc. If ambient air temperature is less than 30 °C (86 °F), cuvette temperatures greater than 35 °C may be difficult to achieve; use the highest achievable temperature.
5. (Manual Option) Set the light flux set-point: ‘Environment’ → ‘Light’ → ‘Head Light Source’. Use 2000, 1600, 1200 $\mu\text{mol m}^{-2} \text{s}^{-1}$, etc.
6. (Manual Option) Wait until all environmental set-points have been reached (green 7/7 badge on the ‘Environment’ tab).
7. (Manual Option) **Run a rapid $A-C_i$ curve**: ‘Programs’ → ‘BP Builder’. Browse to **dynamic/-DAT_CO2_continuous.py**, select it, and click ‘Start BP’. Use the parameters in Table 4.

¹Range matches are temperature sensitive and should be repeated between subsequent completions of the protocol in a field setting.

Table 4: BP Builder parameters for rapid $A-C_i$ program

Parameter	Value
Starting CO ₂	200 $\mu\text{mol mol}^{-1}$
Pre-ramp wait	1 min
Ending CO ₂	2000 $\mu\text{mol mol}^{-1}$
Ramp rate	300 $\mu\text{mol mol}^{-1} \text{ min}^{-1}$
Logging interval	3 s

When done, the instrument will automatically return to 200 $\mu\text{mol m}^{-2} \text{ s}^{-1}$. Select ‘Continue’ to run it. The status ‘1 BP Active’ appears in the upper right. After a 1-min pre-ramp wait, the CO₂ ramp begins; monitor on the ‘BP Monitor’ tab.

When the BP is complete, close (saving) the log file, change light and/or temperature to the next combination, wait for the environment to reach all 7/7 set-points, open a new log file, and repeat.

If using the automated scripts instead of the above manual option, perform the following after selecting and clamping onto a leaf as described above.

1. (Automated Option) Create a folder with the crop name and any desired identifiers.
2. (Automated Option) **Open a log file:** ‘Log Setup’ → ‘Open a Log File’. Use the naming convention: {YYYY-MM-DD}_{time}_light followed by a unique leaf identifier if measuring more than one leaf in this folder.
3. (Automated Option) Run the light-sweep Background Program in ‘Programs’ → ‘BP Builder’. Browse to the location of the downloaded program **ACi.Light.Sweep.py**, select it, and press ‘Start BP’. Use the default settings. Allow the program to run for 40 minutes. Close the log file.
4. (Automated Option) **Open a log file:** ‘Log Setup’ → ‘Open a Log File’. Use the naming convention: {YYYY-MM-DD}_{time}_temperature followed by a unique leaf identifier if measuring more than one leaf in this folder.
5. (Automated Option) Run the temperature-sweep Background Program in ‘Programs’ → ‘BP Builder’. Browse to the location of the downloaded program **ACi.Temperature.Sweep.py**, select it, and press ‘Start BP’. Use the default settings. Allow the program to run for 50 minutes. Close the log file.

5.5 Survey measurement

Table 5: Instrument settings for LI-6800 (LI-COR Biosciences, Lincoln, NE, USA) to be used for survey measurements.

Group	Parameter	Value
Flow	Flow	Auto – 500 $\mu\text{mol s}^{-1}$
	ΔP	Auto – 0.1 kPa
	Pump (speed)	Auto
H ₂ O	RH _{air}	50%
CO ₂	CO _{2_r}	400 $\mu\text{mol mol}^{-1}$
	Soda Lime – Scrub	Auto
Fan	Fan Speed	10 000 rpm
Temperature	Temperature	25 °C
Light	Head Light Source Setpoint	2000 $\mu\text{mol m}^{-2} \text{ s}^{-1}$

1. Perform CO₂ and H₂O point matching as described above in Sect. 5.3.2.
2. **Open a log file:** ‘Log Setup’ → ‘Open a Log File’. Name it {YYYY-MM-DD}_{time}_A_survey and save it in the same folder as the $A-C_i$ curves.

Randomly select 15–25 healthy, sunlit, fully-expanded leaves (as time allows) and collect one measurement per leaf:

1. Set air temperature to $T_{\text{air}} = 25^\circ\text{C}$.
2. Set light flux set-point to $2000 \mu\text{mol m}^{-2} \text{s}^{-1}$.
3. Ensure other environment values match Table 5.
4. Wait until all environmental set-points are met (green 7/7).
5. Choose a healthy, mature, sunlit leaf and clamp the chamber.
6. Wait until the A reading stabilizes (less than 3% deviation from a mean value across 10 seconds; typically after 30 seconds - 2 minutes); record the measurement by pressing the analog ‘Log’ button on the center of the head handle, or by pressing the digital ‘Log’ button in the upper right corner of the console display. Confirmation of a successfully recorded log will be indicated with a chime from the instrument.
7. Move to the next leaf and repeat.

6 Data analysis

6.1 Re-scaling of response curves with survey statistics

A complete sampling of $A-C_i$ curves across several cuvette light and temperature conditions on a single leaf produces the most internally consistent and least error-prone dataset for photosynthetic model parameter extraction. However, leaves in the same plant canopy exhibit variability in photosynthetic parameters primarily due to different nitrogen content as a result of age or developmental plasticity (Niinemets et al., 2004; Xu et al., 2019). To address this issue, we use survey measurements at saturating light and CO₂ and at a reference temperature to re-scale the $A-C_i$ curves using the median saturated A from the survey measurements (Fig. 2). Each set of response curves performed on a single leaf are re-scaled by a factor according to

$$A_{i,\text{scaled}} = A_i \cdot \frac{\text{median}(A_{\text{sat},T_{\text{ref}},\text{survey}})}{A_{\text{sat},T_{\text{ref}},\text{response}}}, \quad (1)$$

for each i^{th} data point of the same leaf, where $A_{\text{sat},T_{\text{ref}},\text{survey}}$ is the set of all surveyed light- and CO₂-saturated A at a constant reference temperature, T_{ref} , of a sample of leaves, and $A_{\text{sat},T_{\text{ref}},\text{response}}$ is the saturated (maximum) value of A obtained in the $A-C_i$ response curve at the same conditions as the survey measurements ($Q_{\text{in}} = 2000 \mu\text{mol m}^{-2} \text{s}^{-1}$, $C_a = 2000 \mu\text{mol mol}^{-1}$, $T_{\text{air}} = T_{\text{ref}}^\circ\text{C}$). Put simply, each group of same-leaf $A-C_i$ curves is shifted up or down so that the end of the $T_{\text{ref}} - Q_{2000}$ curve is aligned with the center of the survey data. T_{ref} used in the example data shown (Figs. 2, 4, and 5) is 25°C .

This approach inherently assumes that leaf-to-leaf variation of photosynthetic responses within a species, cultivar, or genotype exists in the *scale* of the response, while the *shape* is conserved. Evidence demonstrates that *shape* is not entirely conserved (Ogren, 1993), but analysis of Koyama and Kikuzawa (2010) demonstrates that *shape* conservation is a good working approximation.

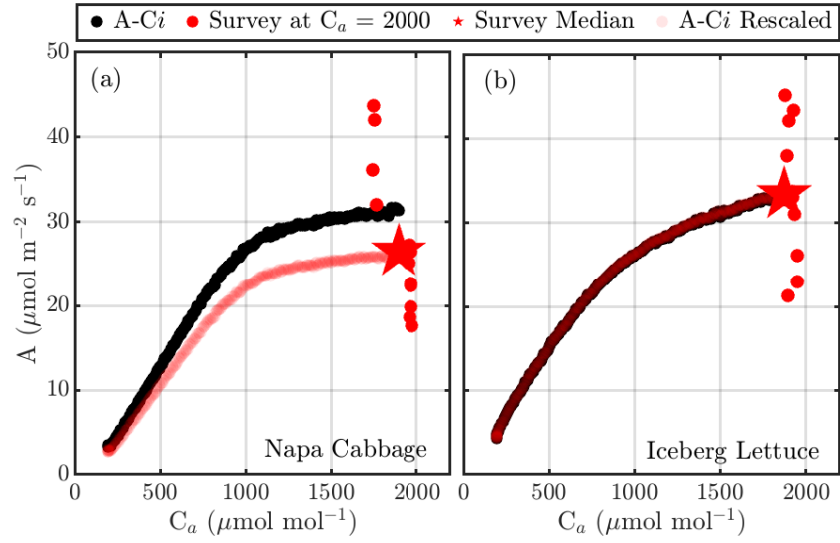


Figure 2: $A-C_i$ curves (black points) of napa cabbage (*Brassica rapa* subsp. *pekinensis*) in (a) and iceberg lettuce (*Lactuca sativa* var. *capitata*) in (b) re-scaled using the approach described in Sect. 6.1 with survey measurements (red points) of ~ 15 randomly sampled healthy, mature leaves at matching cuvette conditions to the $A-C_i$ curves — T_{air} set to 25 °C, Q_{in} set to 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ — and at CO_2 set to 2000 $\mu\text{mol mol}^{-1}$. The median of the survey measurements (red star) were used in the rescaling factor. The leaf chosen for the $A-C_i$ curves in napa cabbage happened to be higher than the median, while the leaf chosen in iceberg lettuce happened to be nearly exactly at the median. Rescaling based on survey measurements helps alleviate the risk that a single leaf chosen for multiple $A-C_i$ is not representative of the canopy median. All subsequent $A-C_i$ curves performed on the same leaf are divided by the same factor.

6.2 Model Fitting

Fitting photosynthetic response curves to the FvCB model, or models of similar complexity, may be conceptually straightforward, but the task has proven unexpectedly challenging in practice (Long and Bernacchi, 2003; Gu et al., 2010; Lochocki and McGrath, 2025), leading to several independently developed fitting solutions. In the R programming language, there are several packages (Liu et al., 2021), including `plantecophys` (Duursma, 2015), `msuRACiFit` (Gregory et al., 2021), `photosynthesis` (Stinziano et al., 2021), `PhotoGEA` (Lochocki et al., 2025). In Python, there is `PhoTorch` (Lei et al., 2025). For this protocol, we demonstrate `PhoTorch Studio` (<https://github.com/ktrizzo/photorch-studio>), a browser-based user interface that packages the robust capabilities of `PhoTorch` into a simple *point-and-click* solution. The interface allows for quick *drag-and-drop* LI-6800 file uploading, seamless rescaling from survey measurements, simple toggling of fitting options including which parameters to keep constant, and plotting of the resulting fit with error metrics. Example results are shown in Figs. 4-6 and Table 6. See Lei et al. (2025) for the full FvCB model description and list of all parameters considered. Detailed instructions for `PhoTorch` and `PhoTorch Studio` can be found in their code repositories.

A key feature of this protocol is its ability to resolve the temperature optima of photosynthetic rates. Component rates of carboxylation, electron transport, and triose-phosphate utilization, V_{cmax} , J_{max} , and TPU , respectively, can exhibit peaked temperature responses of the form

$$k(T) = k_{\text{ref}} \exp \left[\frac{\Delta H_a}{R} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T} \right) \right] \frac{f(T_{\text{ref}})}{f(T)}, \quad (2)$$

$$f(T) = 1 + \exp \left[\frac{\Delta H_d}{R} \left(\frac{1}{T_{\text{opt}}} - \frac{1}{T} \right) - \ln \left(\frac{\Delta H_d}{\Delta H_a} - 1 \right) \right], \quad (3)$$

resulting in an emergent temperature optimum of photosynthesis (Fig. 5) that varies with both light and CO_2 (Fig. 7). This formulation has four parameters, the rate at a reference temperature k_{ref} , the rate's temperature optimum T_{opt} , the activation enthalpy ΔH_a , and the deactivation enthalpy ΔH_d ; R is the ideal gas constant ($0.008314 \text{ kJ mol}^{-1} \text{ K}^{-1}$), and T_{ref} is typically taken to be 298.15 K (25 °C). The six temperatures listed in the protocol were chosen to balance measurement time and confident extraction of these parameters, which facilitates modeling the effective photosynthetic temperature optimum (Fig. 7).

PhoTorch Studio

Photosynthesis Stomatal Conductance Pressure-Volume PROSPECT

Photosynthesis Model Fitting

Upload photosynthesis data files

 Drag and drop files here
Limit 200MB per file

Browse files

	2025-05-20-1004_logdata_T35_Q2000	380.7KB	×
	2025-05-20-1004_logdata_T25_Q1200	392.2KB	×
	2025-05-20-0954_logdata_T25_Q1600	390.2KB	×

Showing page 4 of 5

☐ Skip Header Lines

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✓ Loaded 2161 rows from 14 files.

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Figure 3: PhoTorch Studio, a graphical user interface for the Python-based PhoTorch plant model fitting package. Files directly from the LI-6800, or minimally pre-processed .xlsx or .csv files containing photosynthetic response curve data may be simply dragged and dropped into the app, and fit within seconds. Survey data files may also be included, and optionally used to re-scale response curves. After fitting, results are graphed similarly to Figs. 4-6.

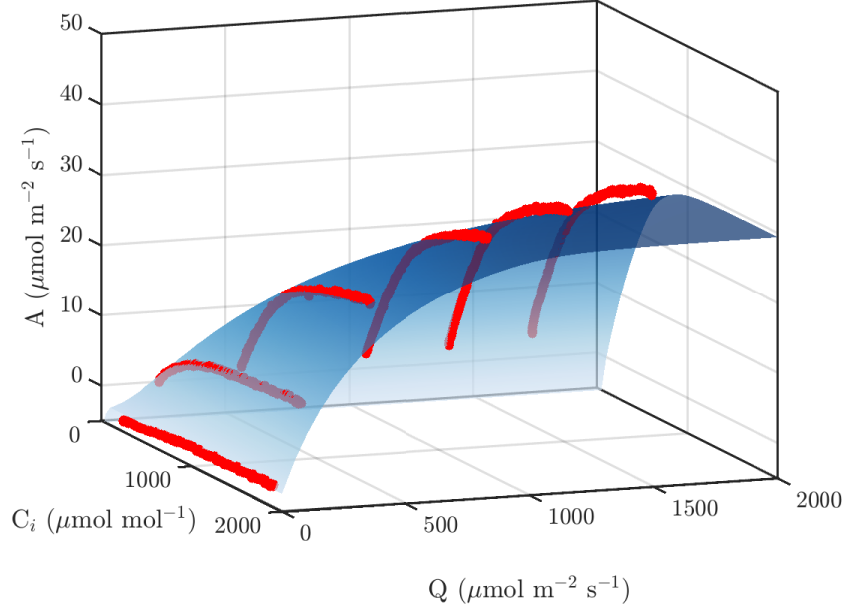


Figure 4: Measured $A-C_i$ curves (red) performed with the Dynamic AssimilationTM technique of the LI-6800 (LI-COR Biosciences, NE, USA) at different light intensities (Q), shown alongside the best fit Farquhar, von Caemmerer, Berry (1980) model surface (blue). The surface is characterized by one consistent set of parameters fit to a number of $A-C_i$ curves at different light intensities and temperatures, and was fit to using the Python package PhoTorch (Lei et al., 2025). The coefficient of determination, R^2 , of the model fit was 0.974.

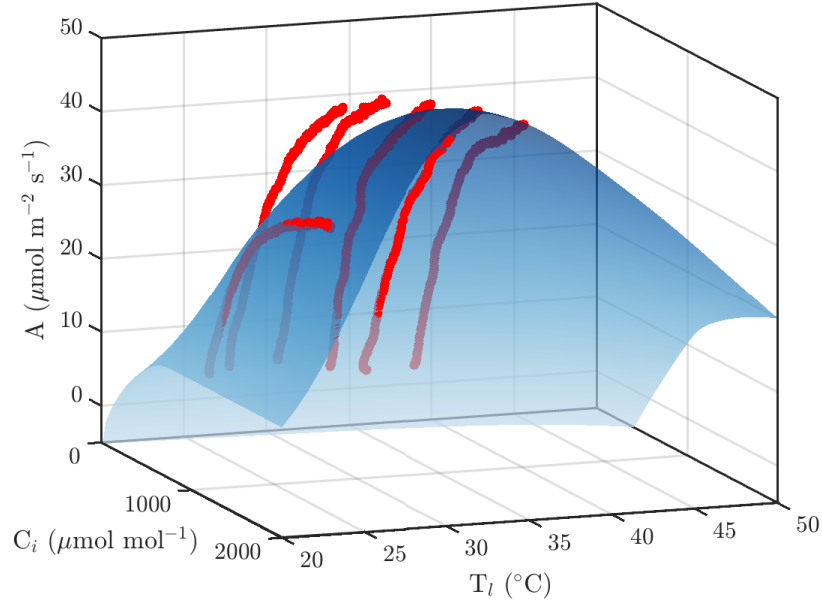


Figure 5: Measured $A-C_i$ curves (red) performed with the Dynamic AssimilationTM technique of the LI-6800 (LI-COR Biosciences, NE, USA) at different leaf temperatures (T_l), shown alongside the best fit Farquhar, von Caemmerer, Berry (1980) model surface (blue). The surface is characterized by one consistent set of parameters fit to a number of $A-C_i$ curves at different light intensities and temperatures, and was fit to using the Python package PhoTorch (Lei et al., 2025). The coefficient of determination, R^2 , of the model fit was 0.974.

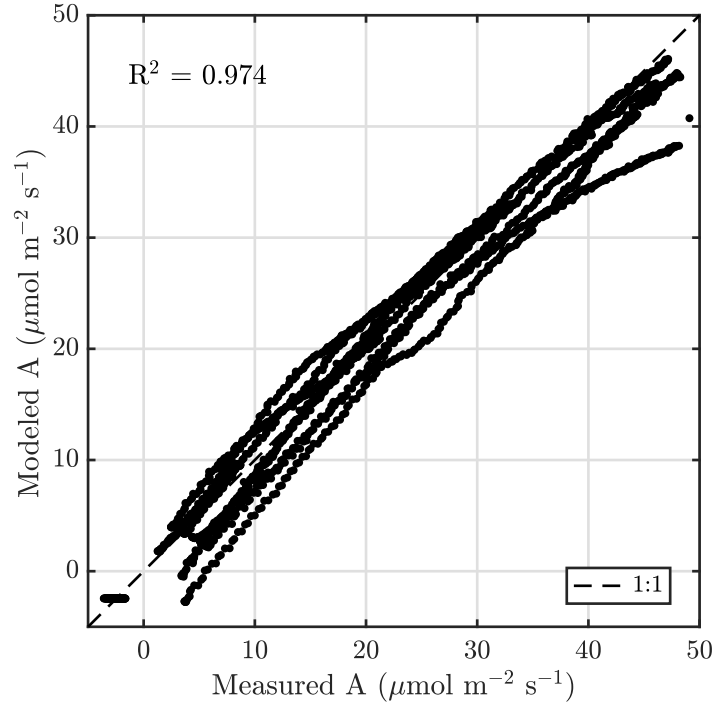


Figure 6: Measured versus modeled net assimilation (A). The measured data consists of $A-C_i$ response curves performed with the Dynamic AssimilationTM technique of the LI-6800 (LI-COR Biosciences, NE, USA) at different light intensities (Q) and leaf temperatures (T_l). Modeled A was obtained by fitting all of the curves together to the Farquhar, von Caemmerer, Berry (1980) model for one set of parameters using the Python package PhoTorch (Lei et al., 2025), and then predicting A given the inputs of C_i , Q , T_l , and the best fit parameters.

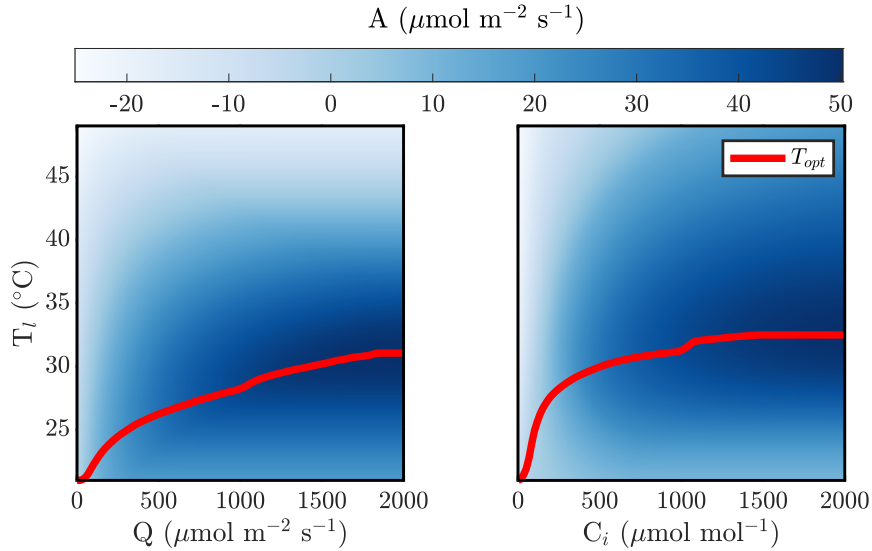


Figure 7: A demonstration of the photosynthetic temperature optimum (T_{opt} ; red) as it varies across leaf temperature (T_l), absorbed photosynthetically active radiation (Q), and internal CO_2 concentration (C_i). Modeled assimilation (A) shown with a colormap was produced using the best-fit Farquhar, von Caemmerer, Berry (1980) model parameters in Table 6, fit from the data in Figs 4 and 5.

Table 6: Extracted Farquhar, von Caemmerer, Berry model parameters of data shown in Figs. 4 and 5, fit with PhoTorch, resulting in a model coefficient of determination, R^2 , of 0.974 (Fig. 6). Input data consisted of 12 Dynamic AssimilationTM Technique A- C_i (LI-6800, LI-COR Biosciences, NE, USA) response curves across a range of cuvette light setpoints from 0-2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ and cuvette air temperatures from 25-40 °C, comprised of 3232 total data points. Response curves varying light were performed on one leaf of an iceberg lettuce (*Lactuca sativa* var. *capitata* ‘Calmar’), and curves varying temperature on another, and scaled using the median of survey measurements (Fig. 2).

Parameter	Value	Fit or Fixed	Description	Units
$V_{\text{cmax}25}$	77.07	fit	max carboxylation rate (Rubisco)	$\mu\text{mol m}^{-2} \text{s}^{-1}$
$J_{\text{max}25}$	191.7	fit	max electron transport rate (RuBP regeneration)	$\mu\text{mol m}^{-2} \text{s}^{-1}$
TPU_{25}	9.607	fit	triose-phosphate utilization rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$
$R_{\text{d}25}$	2.442	fit	mitochondrial respiration rate	$\mu\text{mol m}^{-2} \text{s}^{-1}$
α	0.4365	fit	quantum efficiency (light response initial slope)	unitless
θ	0.1216	fit	light response curvature	–
Γ^*_{25}	72.86	fit	photorespiratory CO_2 compensation point	$\mu\text{mol mol}^{-1}$
$K_{\text{c}25}$	435.3	fit	Michaelis-Menten constant of Rubisco for CO_2	$\mu\text{mol mol}^{-1}$
$K_{\text{o}25}$	238.7	fit	Michaelis-Menten constant of Rubisco for O_2	mmol mol^{-1}
O	213.5	fixed	O_2 concentration	mmol mol^{-1}
g_{m}	0.4232	fit	mesophyll conductance	$\text{mol m}^{-2} \text{s}^{-1}$
α_g	0.000	fit	fraction of photorespiratory glycolate recycling	–
$T_{\text{opt}, V_{\text{cmax}}}$	310.8	fit	optimum temperature of V_{cmax}	K
$T_{\text{opt}, J_{\text{max}}}$	307.4	fit	optimum temperature of J_{max}	K
$T_{\text{opt}, TPU}$	305.5	fit	optimum temperature of TPU	K
$\Delta H_{\text{a}, V_{\text{cmax}}}$	186.7	fit	activation enthalpy of V_{cmax}	kJ mol^{-1}
$\Delta H_{\text{a}, J_{\text{max}}}$	167.0	fit	activation enthalpy of J_{max}	kJ mol^{-1}
$\Delta H_{\text{a}, TPU}$	202.5	fit	activation enthalpy of TPU	kJ mol^{-1}
$\Delta H_{\text{a}, R_{\text{d}}}$	46.39	fit	activation enthalpy of R_{d}	kJ mol^{-1}
$\Delta H_{\text{a}, \Gamma^*}$	37.83	fixed	activation enthalpy of Γ^*	kJ mol^{-1}
$\Delta H_{\text{a}, K_{\text{c}}}$	79.43	fixed	activation enthalpy of K_{c}	kJ mol^{-1}
$\Delta H_{\text{a}, K_{\text{o}}}$	36.38	fixed	activation enthalpy of K_{o}	kJ mol^{-1}
$\Delta H_{\text{d}, V_{\text{cmax}}}$	223.9	fit	deactivation enthalpy of V_{cmax}	kJ mol^{-1}
$\Delta H_{\text{d}, J_{\text{max}}}$	210.1	fit	deactivation enthalpy of J_{max}	kJ mol^{-1}
$\Delta H_{\text{d}, TPU}$	242.5	fit	deactivation enthalpy of TPU	kJ mol^{-1}

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